Algorithms and Computability

Lecture 5: Time Complexity

Martin Zimmermann (Aalborg University)

Last Week in Algorithms and Computability

We have seen

- mapping reductions,
- non-computable problems and how to prove them non-computable via reductions, and
- Rice's theorem: everything "interesting" about Turing machines is not computable.
- Consequence: everything "interesting" about programs is not computable.

Agenda

1. **Motivation**
2. Big-O Notation
3. Complexity Theory
4. Time Complexity

Remember the Entscheidungsproblem?

Lecture 2

The "Entscheidungsproblem" (Hilbert and Ackermann, 1928)

> *Is there an algorithm that, given a statement in some logical language (typically predicate logic), answers "Yes" or "No" according to whether the statement is universally valid.*

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In our language: Is $\{w \mid w \text{ encodes valid statement}\}$ computable?

Lecture 4

The Entscheidungsproblem is not computable.

A Computable Problem

- Presburger Arithmetic (named after Mojżesz Presburger) is a logical system for reasoning about arithmetic using the following axioms:
  - $\forall x.\ \neg(x + 1 = 0)$
  - $\forall x, y.\ (x + 1 = y + 1) \rightarrow x = y$
  - $\forall x.\ x + 0 = x$
  - $\forall x, y.\ x + (y + 1) = (x + y) + 1$
  - $[P(0) \wedge (\forall x.\ P(x) \rightarrow P(x + 1))] \rightarrow \forall x.\ P(x)$ for all Presburger formulas P (the scheme of induction)

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  - $[P(0) \wedge (\forall x.\ P(x) \rightarrow P(x + 1))] \rightarrow \forall x.\ P(x)$ for all Presburger formulas P (the scheme of induction)
- With these axioms you can prove, e.g.,

$$\forall x \exists y.\ (x = y + y) \vee (x = y + y + 1)$$

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  - $[P(0) \wedge (\forall x.\ P(x) \rightarrow P(x + 1))] \rightarrow \forall x.\ P(x)$ for all Presburger formulas P (the scheme of induction)
- The provable statements of Presburger Arithmetic form a computable language **[Presburger '29]**.
- But: Any halting DTM for that language has to move its head at least $2^{2^{cn}}$ times on some inputs of length n (for some constant c) **[Fischer, Rabin '74]**.

Doubly-exponential Growth

Consider $2^{2^{cn}}$ for $c = 1$:

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A Change in Perspective

We studied the following question:

What problems can be solved by a computer?

- The computable problems are those that can be solved by a computer.
- The computably-enumerable problems are those where a "yes"-answer can be verified by a computer.
- Mapping reductions can be used to find relations between problems.

A Change in Perspective

We will study the following question:

What problems can be solved efficiently by a computer?

- The complexity class P contains the problems that can be solved efficiently by a computer.
- The complexity class NP contains the problems where a "yes"-answer can be verified efficiently by a computer.
- Polynomial-time reductions can be used to find relations between problems.

Agenda

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Reminder

Definition

Let $f, g \colon \mathbb{N} \to \mathbb{R}_{>0}$ be functions. We write $f(n) = \mathcal{O}(g(n))$ if

- there are positive integers c, n_0 such that
- $f(n) \leq c \cdot g(n)$ for all $n \geq n_0$.

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Examples

- $2n^2 + 5 = \mathcal{O}(n^2)$ (pick $c = 3$ and $n_0 = 3$)
- $7n^5 + 88n^4 + n^2 + 2013 = \mathcal{O}(n^5)$
- $\log_2(n^7) = \mathcal{O}(\log_2 n)$

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Reading

CLRS Section 3.1 and 3.2 on Big-O notation.

Why Do We Care about Growth Rates?

Assume your computer performs 1 billion steps per second. The table shows the CPU time for inputs of size n.

n	$f(n) = n$	$f(n) = n^2$	$f(n) = n^3$	$f(n) = 2^n$
10	0.01 microsec	0.1 microsec	1 microsec	1 microsec
20	0.02 microsec	0.4 microsec	8 microsec	1 millisec
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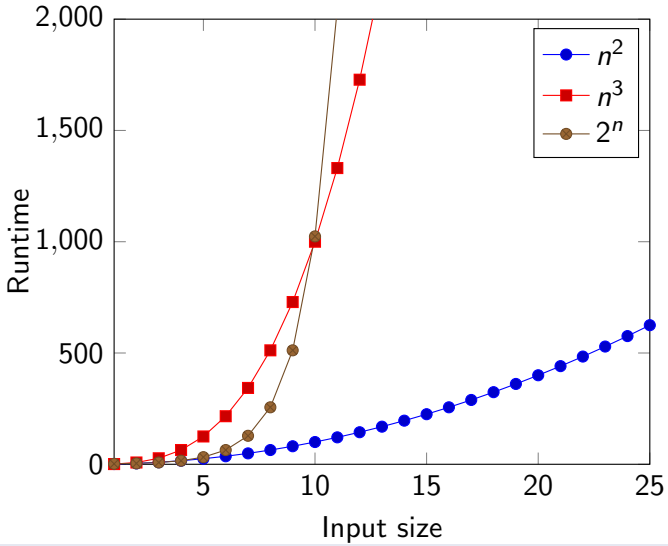

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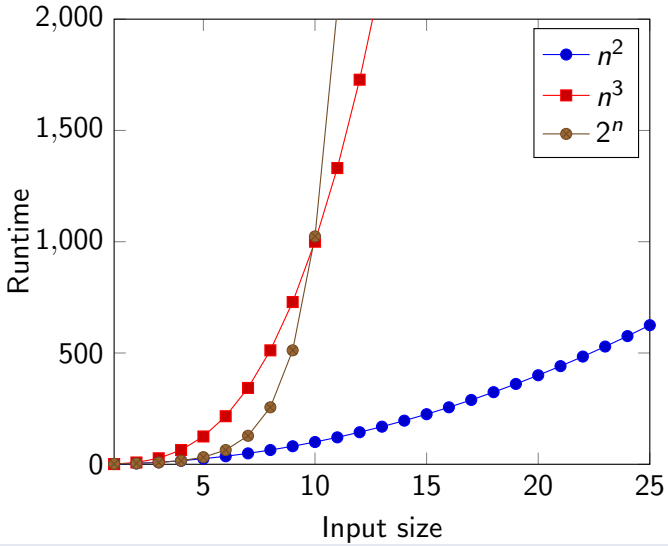

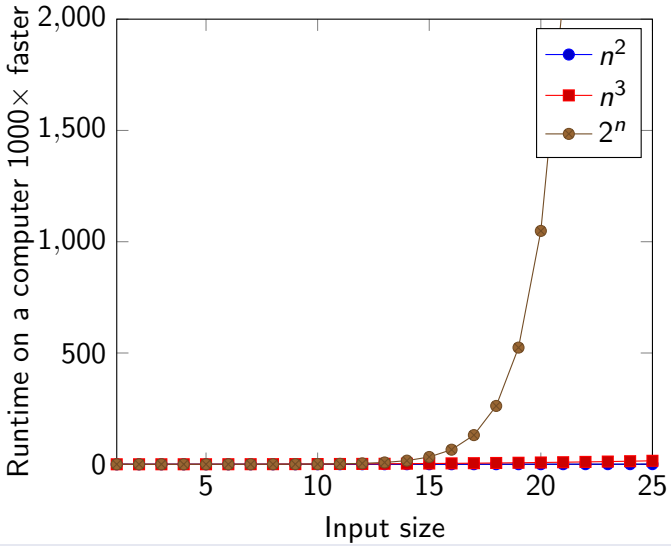

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Intuition

Consider the following halting DTM accepting $\{0^m 1^m \mid m \geq 1\}$:

On input w:

1. Pass once over tape content and reject if it is not of the form 0^+1^+.
2. Pass once over the tape content and replace first 0 by X and first 1 by X (if not possible reject).
3. Pass once over the tape content. If there is still a 0 but no 1 or no 0 but still a 1 then reject. If no 0 and no 1 are left accept.
4. Goto 2.

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How many steps does the Turing machine take on an input of length n at most? $\mathcal{O}(n^2)$ steps overall.

Complexity of Algorithms

On the previous slide, we have analyzed the (worst-case) time-complexity of one halting DTM (algorithm) for $\{0^m 1^m \mid m \geq 1\}$.. but there are many different DTM's!

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Consider the sorting problem:

1. Insertion sort requires n^2 comparisons in the worst case.
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Note

The first two statements are about specific algorithms, the third one is about all (comparison-based) algorithms.

Complexity Theory

Focus of algorithm complexity:

- Study concrete algorithms and their complexity.

Focus of complexity theory:

- Study problems (i.e., languages) instead of algorithms.
- Goal: classify problems according to how easy/hard they are to solve.

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From now on:

- Today, we only consider halting DTM's, i.e., only deterministic Turing machines that halt on every input.
- We analyze the (worst-case) time-complexity of DTM's. Why Turing machines?

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Extended Church–Turing thesis

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- Under (unproven, but widely assumed) complexity-theoretic assumptions: the word "probabilistic" can be dropped.
- Under (unproven, but widely assumed) complexity-theoretic assumptions: The extended Church-Turing thesis is false! Efficient quantum Turing machines cannot be simulated efficiently by (even probabilistic) Turing machines.

Nevertheless, Turing machines are a robust model for computation and can efficiently simulate almost all other models of computation.

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Runtime of a Deterministic Turing Machine

Definition

Let M be a halting DTM.

- Let $time_M(w)$ denote the number of configurations in the unique run of M on input w.
- Let $T\colon \mathbb{N} \to \mathbb{R}_{>0}$. We say that M runs within time T if $time_M(w) \leq T(|w|)$ for all inputs w.

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We are interested in classifying problems (languages) according to their asymptotic time-complexity.

Definition

Let $T\colon \mathbb{N} \to \mathbb{R}_{>0}$ be a function. The complexity class $\textsc{Time}(T)$ is defined as

$$\textsc{Time}(T) = \{L(M) \mid M \text{ is a halting DTM that runs within time } \mathcal{O}(T)\}.$$

Time Hierarchy

A consequence of a more general theorem:

$$\textsc{Time}(n) \subsetneq \textsc{Time}(n^2) \subsetneq \textsc{Time}(n^3) \subsetneq \textsc{Time}(n^4) \subsetneq \cdots \subsetneq \textsc{Time}(2^n)$$

Intuitively: more time allows you to compute more languages.

Proof: via diagonalization construct a language that is different from every language in $\textsc{Time}(n^k)$, but in $\textsc{Time}(n^{k+1})$. Note that these languages are all artificial.

Robustness

For one-tape halting DTM's:

- $\{0^m 1^{m'} \mid m, m' \geq 1\} \in \textsc{Time}(n)$
- $\{0^m 1^m \mid m \geq 1\} \in \textsc{Time}(n^2)$
- $\{w\#w \mid w \in \mathbb{B}^*\} \in \textsc{Time}(n^2)$

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In general:

Theorem

Let $t\colon \mathbb{N} \to \mathbb{R}_{>0}$ be a function such that $t(n) \geq n$. Every k-tape halting DTM with time-complexity $t(n)$ can be simulated by an equivalent one-tape halting DTM with time-complexity $t^2(n)$.

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Proof: Simulation presented in Lecture 2 has the desired properties.

The Complexity Class P

Definition

The complexity class P (polynomial time) is defined as

$$\mathrm{P} = \bigcup_{k \geq 0} \textsc{Time}(n^k)$$

- Robust definition (can use other deterministic (!) models of computation, e.g., multi-tape Turing machines).
- Cobham's thesis: A problem can be efficiently computed if and only if it is in P.

About Cobham's Thesis

A problem with fastest algorithm with running time..

- n^{58}: efficient!
- $n^{\log_2 \log_2 \log_2 \log_2 n}$: not efficient!

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- $n^{\log_2 \log_2 \log_2 \log_2 n}$: not efficient! But it is roughly $3n$ for $n = 1.000.000$ and roughly $60n$ for $n = 1.000.000.000$.
- In practical applications: Even quadratic running time infeasible for data-intensive problems.

Problems in P

Many problems are in P:

- $\{0^m 1^{m'} \mid m, m' \geq 1\} \in \textsc{Time}(n)$
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- graph reachability
- NFA emptiness
- Primality
- Linear Programming
- Word problem for contextfree grammars
- Matchings
- .. and many other problems

Closure Properties

For reference:

Theorem

P *is closed under union, intersection, complementation, concatenation, and iteration.*

Proof: The constructions presented in Lecture 3 and Tutorial 3 yield halting DTM's with polynomial time-complexity, when applied to halting DTM's with polynomial time-complexity.

Conclusion

We have compared algorithmic complexity:

- study of concrete algorithms and precise running times
- difference between $\mathcal{O}(n^2)$ and $\mathcal{O}(n^3)$ is huge
- running times depend on model of computation

and complexity theory:

- study of problems (languages) rather than algorithms
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Note

Algorithmic complexity is still important for complexity theory: giving an algorithm with polynomial running time for a problem L implies $L \in \mathrm{P}$.

Reading

Wait till next lecture!